Mat2002 Tutorial 2

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Quick review

- Normal form: $y' = f(x)$
- Separable equation: $h(y)dy = g(x)dx$

  Constant solution!
- Linear equation: $y' + p(x)y = q(x)$

$$\mu = e^{\int p(x)dx}$$

Today

1. Substitution:
(1) linear substitution: $\frac{dy}{dx} = f(ax + by + c)$
(2) Homogeneous equation: $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$
(3) Bernoulli Equation: $y' + p(x)y = y^n$

2. (1) Exact equation: $Mdy + Ndx = 0$.
(2) Non- exact equation can be made exact.

Linear substitution

Example

$\frac{dy}{dx} = (x + y + 3)^2$
Sol:Let $v = x + y + 3$ then

$$\frac{dv}{dx} = \frac{dy}{dx} + 1$$

Thus: $\frac{dv}{dx} = v^2 + 1$

$$\frac{dv}{v^2 + 1} = dx$$
$$\arctan v = x + c$$
$$v = \tan(x + c)$$
$$y = \tan(x + c) - x - 3$$

Homogeneous equation

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right) : \text{ Let } \quad v = \frac{y}{x} \quad \frac{dy}{dx} = x \cdot \frac{dv}{dx} + v$$

Example

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

Sol: $\frac{dy}{dx} = \frac{1+\frac{y}{x}}{1-\frac{y}{x}} = \frac{1+v}{1-v}$

$x\frac{dv}{dx} = \frac{1+v}{1-v} - v = \frac{1+v^2}{1-v}$

$\frac{1-v}{1+v^2}dv = \frac{1}{x}dx$

$\arctan v - \ln\sqrt{1+v^2} = \ln|x| + C$

Substitute back: $\sqrt{x^2+y^2} = ce^{\arctan\frac{y}{x}} (c > 0)$

Bernoulli equation

$$y' + p(x)y = q(x)y^n$$

(i) $n = 0.1$ this is linear.
(ii) $n \neq 0, 1$. one solution is $y \equiv 0$!
For $y \neq 0$. Let $z = y^{1-n}$. Then: $\frac{dz}{dx} = (1-n)y^{-n} \cdot y'$

$$= (1-n)y^{-n} \cdot (-p(x)y + q(x)y^n)$$
$$= -(1-n)p(x)z + (1-n)q(x)$$

This is a First order ODE for z.

Bernoulli equation

Example

$$y' + \frac{2}{x}y = \frac{1}{x^2}y^3 \quad x > 0$$

(i) $y = 0$ is a solution

(ii) $y \neq 0$. Let $z = y^{-2}$, then $z' = -2y^{-3} \cdot y' = -2y^{-3}\left(-\frac{2}{x}y + \frac{1}{x^2}y^3\right)$

$z' - \frac{4}{x}z = -\frac{2}{x^2}$

$\mu = x^{-4}$

$z = \frac{2}{5x} + cx^4$

$y = \pm\sqrt{\frac{5x}{2+cx^5}}$

Exercise

Exercise

$y' = \frac{1}{x+y}$

$y' = \frac{y-4x}{x-y}$

$y' = x^3y^3 - xy$

Exercise

Solution

(i) $y' = \frac{1}{x+y}$

$$y = -x - 1 (for \quad v = -1)$$
$$y - ln|x + y + 1| = C$$

(ii) $y' = \frac{y-4x}{x-y}$

$$x\frac{dv}{dx} = \frac{v-4}{1-v} - v = \frac{v^2-4}{1-v}$$
$$\frac{1-v}{v^2-4}dv = \frac{1}{x}dx$$

Integrate and substitute back

$$-\frac{1}{4}ln|2-v| - \frac{3}{4}ln|2+v| = ln|x| + C$$

Excercise

Solution

$$y' = x^3y^3 - xy$$

Let $z = y^{-2}$

$$z' - 2xz = -2x^3$$

$$\mu = e^{-x^2}$$

$$z = 1 + x^2 + ce^{x^2}$$

$$y = \pm\sqrt{\frac{1}{1 + x^2 + Ce^{x^2}}}$$

Exact Equation

Motivation: If we have $\psi(x, y) = C$. For example $x^2 + xy^3 = C$. When we find y'. we take $\frac{d}{dx}$: $2x + y^3 + 3xy^2y' = 0$
Thus: $\left(2x + y^3\right) dx + \left(3xy^2\right) dy = 0$
$x^2 + xy^3 = C$ is a solution to the above ODE.

How to check: $Mdx + Ndy = 0$ is from taking $\frac{d}{dx}$ to some $\psi(x, y) = 0$ or not?
Theorem: $f_{xy} = f_{yx}$ hence we must have $M_y = N_x$. We say the equation is exact if it satisfied this condition.

So if we have $Mdy + Ndx = 0$ is exact.Then its solution should be some

$$\psi(x, y) = C$$

Exact equation

Example

$$y\cos x + 2xe^y + \left(\sin x + x^2e^y - 1\right)y' = 0$$

$$M = y\cos x + 2xe^y \qquad M_y = \cos x + 2xe^y$$
$$N = \sin x + x^2e^y - 1. \qquad N_x = \cos x + 2xe^y$$

$M_y = N_x$. exact. Let its solution be $\psi(x, y) = C$.

Solve $\begin{cases} \psi_x = M = y\cos x + 2xe^y \\ \psi_y = N = \sin x + x^2e^y - 1. \end{cases}$ It is a PDE. but it is easy:

$\Psi_x = M$ implies
$\psi(x, y) = \int M dx = \int (y\cos x + 2xe^y)\, dx = y\sin x + x^2e^y + h(y)$ (No need to add C here)
$\Psi_y = N$ implies $\sin x + x^2e^y + h'(y) = N$
$h'(y) = -1 \quad h(y) = -y$(No need to add C here)
Solution is: $y\sin x + x^2e^y - x = C$ (Do not miss C here)

Non-Exact equation that can be made exact

Idea: Integrate factor.
$Mdx + Ndy = 0$. may not exact but $(\mu M)dx + (\mu N)dy = 0$ may be exact.
Then we should make μ sit.

$$(\mu M)_y = (\mu N)_x.$$
$$\Leftrightarrow M\mu_y - N\mu_x = \mu\,(N_x - M_y)$$

This is a PDE. we can only solve it for some special cases.
(1) If $\mu = \mu(x) : -N\mu_x = \mu\,(N_x - M_y)$

$$\mu_x = \frac{M_y - N_x}{N}\mu$$

If $\frac{M_y - N_x}{N}$ only depends on x.

$$\mu = e^{\int \frac{M_y - N_x}{N} dx}$$

(2) If $\mu = \mu(y)$: similarly $\mu_y = \frac{N_x - M_y}{M}\mu(y)$ If $\frac{N_x - M_y}{M}$ only depends on y.

$$\mu = e^{\int \frac{N_x - M_y}{M} dy}$$

Non-Exact equation that can be made exact

$(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0$

$$M = 3xy + y^2 \Rightarrow M_y = 3x + 2y$$
$$N = x^2 + xy = N_x = 2x + y$$

Not exact. But $\frac{M_y - N_x}{N} = \frac{x+y}{x^2+xy} = \frac{1}{x}$ only depend on x.

$$\mu = e^{\int \frac{1}{x} dx} = x$$

Matiply by μ : $(3x^2y + xy^2)\,dx + (x^3 + x^2y)\,dy = 0$

$$\bar{M} = 3x^2y + xy^2$$
$$\bar{N} = x^3 + x^2y$$
$$\psi(x, y) = \int M dx = x^3y + \frac{1}{2}x^2y^2 + h(y)$$
$$\psi_y = x^3 + x^2y + h'(y) = N \Rightarrow h'(y) = 0$$

Solution is: $x^3y + \frac{1}{2}x^2y^2 = C$

Exercise

$$2xy^3dx + \left(x^2y^2 - 1\right)dy = 0$$

Exercise

$$2xy^3dx + \left(x^2y^2 - 1\right)dy = 0$$

Answer: $\mu = y^{-2}$
Solution is $= x^2y + \frac{1}{y} = C$.